$$C_6H_5\text{-}\overset{+}{N}_2Cl^- + KI \rightarrow \underset{\text{Iodobenzene}}{C_6H_5I} + KCl + N_2$$

4. Replacement of fluorine (Baltz – schiemann reaction)

When benzene diazonium chloride is treated with fluoroboric acid, benezene diazonium tetra fluoroborate is precipitated which on heating decomposes to give fluorobenzene.

$$\underset{\substack{\text{Fluoroboric}\\ \text{acid}}}{C_6H_5\text{-}\overset{+}{N}_2Cl^- + HBF_4} \longrightarrow \underset{\substack{\text{Benzenediazonium}\\ \text{fluoroborate}}}{C_6H_5\text{-}\overset{+}{N}_2BF_4^-} \xrightarrow{\Delta} \underset{\text{Fluorobenzene}}{C_6H_5\text{-}F} + BF_3 + N_2$$

5. Replacement by hydroxyl group

Benzene diazonium chloride solution is added slowly to a large volume of boiling water to get phenol.

$$C_6H_5\text{-}N_2Cl + H_2O \xrightarrow[283K]{\Delta} \underset{\text{Phenol}}{C_6H_5\text{-}OH} + N_2 + HCl$$

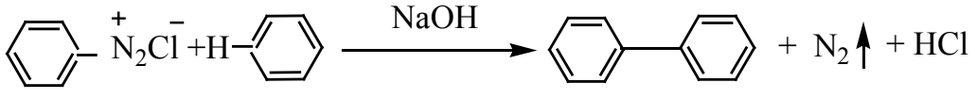
6. Replacement by nitrogroup

When diazonium fluoroborate is heated with aqueous sodium nitrite solution in the presence of copper, the diazonium group is replaced by $-NO_2$ group.

$$\underset{\substack{\text{Fluoroboric}\\ \text{acid}}}{C_6H_5\text{-}\overset{+}{N}_2Cl^- + HBF_4} \longrightarrow C_6H_5\text{-}\overset{+}{N}_2BF_4^- \xrightarrow[Cu,\ \Delta]{NaNO_2} \underset{\text{Nitrobenzene}}{C_6H_5\text{-}NO_2} + N_2 + NaBF_4$$

7. Replacement by aryl group (Gomberg reaction)

Benzene diazonium chloride reacts with benzene in the presence of sodium hydroxide to give biphenyl. This reaction in known as the Gomberg reaction.



Benzene — Biphenyl

8. Replacement by carboxylic acid group

When diazonium fluoroborate is heated with acetic acid, benzoic acid is obtained. This reaction is used to convert the of aliphatic carboxylic acid into aromatic carboxylic acid.

$$\underset{\substack{\text{Benzene diazonium}\\ \text{fluoroborate}}}{C_6H_5\text{-}N_2BF_4} + CH_3\text{-}COOH \longrightarrow \underset{\text{Benzoic acid}}{C_6H_5\text{-}COOH} + BF_3 + CH_3F + N_2$$

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